TAGE- ./sim-outorder -max:inst 500000000 -bpred tage -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000002 # total number of instructions committed
sim_num_refs            219903438 # total number of loads and stores committed
sim_num_loads            67608641 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time              138 # total simulation time in seconds
sim_inst_rate        3623188.4203 # simulation speed (in insts/sec)
sim_total_insn          521967502 # total number of instructions executed
sim_total_refs          225768258 # total number of loads and stores executed
sim_total_loads          72098810 # total number of loads executed
sim_total_stores      153669448.0000 # total number of stores executed
sim_total_branches       53483332 # total number of branches executed
sim_cycle               395925485 # total simulation time in cycles
sim_IPC                    1.2629 # instructions per cycle
sim_CPI                    0.7919 # cycles per instruction
sim_exec_BW                1.3183 # total instructions (mis-spec + committed) per cycle
sim_IPB                    9.8578 # instruction per branch
IFQ_count               839503040 # cumulative IFQ occupancy
IFQ_fcount              197694540 # cumulative IFQ full count
ifq_occupancy              2.1204 # avg IFQ occupancy (insn's)
ifq_rate                   1.3183 # avg IFQ dispatch rate (insn/cycle)
ifq_latency                1.6083 # avg IFQ occupant latency (cycle's)
ifq_full                   0.4993 # fraction of time (cycle's) IFQ was full
RUU_count              3024271359 # cumulative RUU occupancy
RUU_fcount              82238431 # cumulative RUU full count
ruu_occupancy              7.6385 # avg RUU occupancy (insn's)
ruu_rate                   1.3183 # avg RUU dispatch rate (insn/cycle)
ruu_latency                5.7940 # avg RUU occupant latency (cycle's)
ruu_full                   0.2077 # fraction of time (cycle's) RUU was full
LSQ_count              1252822746 # cumulative LSQ occupancy
LSQ_fcount              72711406 # cumulative LSQ full count
lsq_occupancy              3.1643 # avg LSQ occupancy (insn's)
lsq_rate                   1.3183 # avg LSQ dispatch rate (insn/cycle)
lsq_latency                2.4002 # avg LSQ occupant latency (cycle's)
lsq_full                   0.1836 # fraction of time (cycle's) LSQ was full
sim_slip               4882512041 # total number of slip cycles
avg_sim_slip               9.7650 # the average slip between issue and retirement
bpred_tage.lookups         53932507 # total number of bpred lookups
bpred_tage.updates         50721275 # total number of updates
bpred_tage.addr_hits       46684668 # total number of address-predicted hits
bpred_tage.dir_hits        47082877 # total number of direction-predicted hits (includes addr-hits)
bpred_tage.misses           3638398 # total number of misses
bpred_tage.jr_hits          1144674 # total number of address-predicted hits for JR's
bpred_tage.jr_seen          1528398 # total number of JR's seen
```

```
bpred_tage.jr_non_ras_hits.PP       106459 # total number of address-predicted hits for non-RAS JR's
bpred_tage.jr_non_ras_seen.PP       389523 # total number of non-RAS JR's seen
bpred_tage.bpred_addr_rate    0.9204 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_tage.bpred_dir_rate    0.9283 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_tage.bpred_jr_rate    0.7489 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_tage.bpred_jr_non_ras_rate.PP    0.2733 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_tage.retstack_pushes      1195501 # total number of address pushed onto ret-addr stack
bpred_tage.retstack_pops      1376094 # total number of address popped off of ret-addr stack
bpred_tage.used_ras.PP      1138875 # total number of RAS predictions used
bpred_tage.ras_hits.PP      1038215 # total number of RAS hits
bpred_tage.ras_rate.PP    0.9116 # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            548225831 # total number of accesses
il1.hits                528821776 # total number of hits
il1.misses               19404055 # total number of misses
il1.replacements         19403543 # total number of replacements
il1.writebacks                 0 # total number of writebacks
il1.invalidations              0 # total number of invalidations
il1.miss_rate             0.0354 # miss rate (i.e., misses/ref)
il1.repl_rate             0.0354 # replacement rate (i.e., repls/ref)
il1.wb_rate               0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses            215681442 # total number of accesses
dl1.hits                198634933 # total number of hits
dl1.misses               17046509 # total number of misses
dl1.replacements         17045997 # total number of replacements
dl1.writebacks           16837642 # total number of writebacks
dl1.invalidations              0 # total number of invalidations
dl1.miss_rate             0.0790 # miss rate (i.e., misses/ref)
dl1.repl_rate             0.0790 # replacement rate (i.e., repls/ref)
dl1.wb_rate               0.0781 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses             53288206 # total number of accesses
ul2.hits                 44911007 # total number of hits
ul2.misses                8377199 # total number of misses
ul2.replacements          8373103 # total number of replacements
ul2.writebacks            8340513 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate             0.1572 # miss rate (i.e., misses/ref)
ul2.repl_rate             0.1571 # replacement rate (i.e., repls/ref)
ul2.wb_rate               0.1565 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses           548225831 # total number of accesses
itlb.hits               548191071 # total number of hits
itlb.misses                 34760 # total number of misses
itlb.replacements           34696 # total number of replacements
itlb.writebacks                0 # total number of writebacks
itlb.invalidations             0 # total number of invalidations
itlb.miss_rate            0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate            0.0001 # replacement rate (i.e., repls/ref)
```

```
itlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses           222518070 # total number of accesses
dtlb.hits               222347115 # total number of hits
dtlb.misses               170955 # total number of misses
dtlb.replacements         170827 # total number of replacements
dtlb.writebacks                0 # total number of writebacks
dtlb.invalidations             0 # total number of invalidations
dtlb.miss_rate            0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs              0 # total non-speculative bogus addresses seen (debug var)
ld_text_base           0x00400000 # program text (code) segment base
ld_text_size              452000 # program text (code) size in bytes
ld_data_base           0x10000000 # program initialized data segment base
ld_data_size             2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base          0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size              16384 # program initial stack size
ld_prog_entry          0x00400140 # program entry point (initial PC)
ld_environ_base        0x7fff8000 # program environment base address address
ld_target_big_endian           0 # target executable endian-ness, non-zero if big endian
mem.page_count             45075 # total number of pages allocated
mem.page_mem              180300k # total size of memory pages allocated
mem.ptab_misses            93443 # total first level page table misses
mem.ptab_accesses       3961324962 # total page table accesses
mem.ptab_miss_rate         0.0000 # first level page table miss rate
```

Perceptron- ./sim-outorder -max:inst 500000000 -bpred percept -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000003 # total number of instructions committed
sim_num_refs            219903439 # total number of loads and stores committed
sim_num_loads            67608642 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time              152 # total simulation time in seconds
sim_inst_rate         3289473.7039 # simulation speed (in insts/sec)
sim_total_insn          535893682 # total number of instructions executed
```

```
sim_total_refs          228092438 # total number of loads and stores executed
sim_total_loads          74152054 # total number of loads executed
sim_total_stores       153940384.0000 # total number of stores executed
sim_total_branches        53976811 # total number of branches executed
sim_cycle              1003575527 # total simulation time in cycles
sim_IPC                   0.4982 # instructions per cycle
sim_CPI                   2.0072 # cycles per instruction
sim_exec_BW               0.5340 # total instructions (mis-spec + committed) per cycle
sim_IPB                   9.8578 # instruction per branch
IFQ_count              718865038 # cumulative IFQ occupancy
IFQ_fcount             167266425 # cumulative IFQ full count
ifq_occupancy             0.7163 # avg IFQ occupancy (insn's)
ifq_rate                  0.5340 # avg IFQ dispatch rate (insn/cycle)
ifq_latency               1.3414 # avg IFQ occupant latency (cycle's)
ifq_full                  0.1667 # fraction of time (cycle's) IFQ was full
RUU_count             2834713192 # cumulative RUU occupancy
RUU_fcount             43794890 # cumulative RUU full count
ruu_occupancy             2.8246 # avg RUU occupancy (insn's)
ruu_rate                  0.5340 # avg RUU dispatch rate (insn/cycle)
ruu_latency               5.2897 # avg RUU occupant latency (cycle's)
ruu_full                  0.0436 # fraction of time (cycle's) RUU was full
LSQ_count             1138605741 # cumulative LSQ occupancy
LSQ_fcount             40398636 # cumulative LSQ full count
lsq_occupancy             1.1345 # avg LSQ occupancy (insn's)
lsq_rate                  0.5340 # avg LSQ dispatch rate (insn/cycle)
lsq_latency               2.1247 # avg LSQ occupant latency (cycle's)
lsq_full                  0.0403 # fraction of time (cycle's) LSQ was full
sim_slip               4537037577 # total number of slip cycles
avg_sim_slip              9.0741 # the average slip between issue and retirement
bpred_perceptron.lookups     54648346 # total number of bpred lookups
bpred_perceptron.updates     50721278 # total number of updates
bpred_perceptron.addr_hits      5249404 # total number of address-predicted hits
bpred_perceptron.dir_hits     40395198 # total number of direction-predicted hits (includes addr-hits)
bpred_perceptron.misses      10326080 # total number of misses
bpred_perceptron.jr_hits       977797 # total number of address-predicted hits for JR's
bpred_perceptron.jr_seen      1528398 # total number of JR's seen
bpred_perceptron.jr_non_ras_hits.PP           0 # total number of address-predicted hits for non-RAS JR's
bpred_perceptron.jr_non_ras_seen.PP       389523 # total number of non-RAS JR's seen
bpred_perceptron.bpred_addr_rate    0.1035 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_perceptron.bpred_dir_rate    0.7964 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_perceptron.bpred_jr_rate    0.6398 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_perceptron.bpred_jr_non_ras_rate.PP    0.0000 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_perceptron.retstack_pushes      1363128 # total number of address pushed onto ret-addr stack
bpred_perceptron.retstack_pops      1148194 # total number of address popped off of ret-addr stack
bpred_perceptron.used_ras.PP      1138875 # total number of RAS predictions used
bpred_perceptron.ras_hits.PP       977797 # total number of RAS hits
bpred_perceptron.ras_rate.PP    0.8586 # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            547298487 # total number of accesses
il1.hits                527163840 # total number of hits
```

```
il1.misses              20134647 # total number of misses
il1.replacements        20134135 # total number of replacements
il1.writebacks                 0 # total number of writebacks
il1.invalidations              0 # total number of invalidations
il1.miss_rate             0.0368 # miss rate (i.e., misses/ref)
il1.repl_rate             0.0368 # replacement rate (i.e., repls/ref)
il1.wb_rate               0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses           222548387 # total number of accesses
dl1.hits               205501910 # total number of hits
dl1.misses              17046477 # total number of misses
dl1.replacements        17045965 # total number of replacements
dl1.writebacks          16837639 # total number of writebacks
dl1.invalidations              0 # total number of invalidations
dl1.miss_rate             0.0766 # miss rate (i.e., misses/ref)
dl1.repl_rate             0.0766 # replacement rate (i.e., repls/ref)
dl1.wb_rate               0.0757 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses            54018763 # total number of accesses
ul2.hits                45640687 # total number of hits
ul2.misses               8378076 # total number of misses
ul2.replacements         8373980 # total number of replacements
ul2.writebacks           8340728 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate             0.1551 # miss rate (i.e., misses/ref)
ul2.repl_rate             0.1550 # replacement rate (i.e., repls/ref)
ul2.wb_rate               0.1544 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses          547298487 # total number of accesses
itlb.hits              547263909 # total number of hits
itlb.misses                34578 # total number of misses
itlb.replacements          34514 # total number of replacements
itlb.writebacks                0 # total number of writebacks
itlb.invalidations             0 # total number of invalidations
itlb.miss_rate            0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate            0.0001 # replacement rate (i.e., repls/ref)
itlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses          224304503 # total number of accesses
dtlb.hits              224133719 # total number of hits
dtlb.misses               170784 # total number of misses
dtlb.replacements         170656 # total number of replacements
dtlb.writebacks                0 # total number of writebacks
dtlb.invalidations             0 # total number of invalidations
dtlb.miss_rate            0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs              0 # total non-speculative bogus addresses seen (debug var)
```

```
ld_text_base          0x00400000 # program text (code) segment base
ld_text_size             452000 # program text (code) size in bytes
ld_data_base          0x10000000 # program initialized data segment base
ld_data_size            2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base         0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size             16384 # program initial stack size
ld_prog_entry         0x00400140 # program entry point (initial PC)
ld_environ_base       0x7fff8000 # program environment base address address
ld_target_big_endian           0 # target executable endian-ness, non-zero if big endian
mem.page_count            45075 # total number of pages allocated
mem.page_mem             180300k # total size of memory pages allocated
mem.ptab_misses           91856 # total first level page table misses
mem.ptab_accesses      3962314303 # total page table accesses
mem.ptab_miss_rate        0.0000 # first level page table miss rate
```

Agree - ./sim-outorder -max:inst 500000000 -bpred agree -bpred:agree 1 8192 8 1 -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn           500000002 # total number of instructions committed
sim_num_refs           219903438 # total number of loads and stores committed
sim_num_loads           67608641 # total number of loads committed
sim_num_stores       152294797.0000 # total number of stores committed
sim_num_branches        50721278 # total number of branches committed
sim_elapsed_time              131 # total simulation time in seconds
sim_inst_rate        3816793.9084 # simulation speed (in insts/sec)
sim_total_insn         530825719 # total number of instructions executed
sim_total_refs         228496078 # total number of loads and stores executed
sim_total_loads         74215618 # total number of loads executed
sim_total_stores     154280460.0000 # total number of stores executed
sim_total_branches      55423779 # total number of branches executed
sim_cycle              420382106 # total simulation time in cycles
sim_IPC                   1.1894 # instructions per cycle
sim_CPI                   0.8408 # cycles per instruction
sim_exec_BW               1.2627 # total instructions (mis-spec + committed) per cycle
sim_IPB                   9.8578 # instruction per branch
IFQ_count              850318706 # cumulative IFQ occupancy
IFQ_fcount             200025344 # cumulative IFQ full count
ifq_occupancy             2.0227 # avg IFQ occupancy (insn's)
ifq_rate                  1.2627 # avg IFQ dispatch rate (insn/cycle)
ifq_latency               1.6019 # avg IFQ occupant latency (cycle's)
ifq_full                  0.4758 # fraction of time (cycle's) IFQ was full
RUU_count             3055911701 # cumulative RUU occupancy
RUU_fcount             81011811 # cumulative RUU full count
ruu_occupancy             7.2694 # avg RUU occupancy (insn's)
```

```
ruu_rate                1.2627 # avg RUU dispatch rate (insn/cycle)
ruu_latency             5.7569 # avg RUU occupant latency (cycle's)
ruu_full                0.1927 # fraction of time (cycle's) RUU was full
LSQ_count               1264605939 # cumulative LSQ occupancy
LSQ_fcount              72783983 # cumulative LSQ full count
lsq_occupancy           3.0082 # avg LSQ occupancy (insn's)
lsq_rate                1.2627 # avg LSQ dispatch rate (insn/cycle)
lsq_latency             2.3823 # avg LSQ occupant latency (cycle's)
lsq_full                0.1731 # fraction of time (cycle's) LSQ was full
sim_slip                4872734243 # total number of slip cycles
avg_sim_slip            9.7455 # the average slip between issue and retirement
bpred_agree.lookups     56144335 # total number of bpred lookups
bpred_agree.updates     50721275 # total number of updates
bpred_agree.addr_hits   45465575 # total number of address-predicted hits
bpred_agree.dir_hits    45865416 # total number of direction-predicted hits (includes addr-hits)
bpred_agree.misses      4855859 # total number of misses
bpred_agree.jr_hits     1140326 # total number of address-predicted hits for JR's
bpred_agree.jr_seen     1528398 # total number of JR's seen
bpred_agree.jr_non_ras_hits.PP     106459 # total number of address-predicted hits for non-RAS JR's
bpred_agree.jr_non_ras_seen.PP     389523 # total number of non-RAS JR's seen
bpred_agree.bpred_addr_rate    0.8964 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_agree.bpred_dir_rate     0.9043 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_agree.bpred_jr_rate      0.7461 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_agree.bpred_jr_non_ras_rate.PP    0.2733 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_agree.retstack_pushes    1371400 # total number of address pushed onto ret-addr stack
bpred_agree.retstack_pops      1223685 # total number of address popped off of ret-addr stack
bpred_agree.used_ras.PP        1138875 # total number of RAS predictions used
bpred_agree.ras_hits.PP        1033867 # total number of RAS hits
bpred_agree.ras_rate.PP        0.9078 # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            560197796 # total number of accesses
il1.hits                540185444 # total number of hits
il1.misses              20012352 # total number of misses
il1.replacements        20011840 # total number of replacements
il1.writebacks          0 # total number of writebacks
il1.invalidations       0 # total number of invalidations
il1.miss_rate           0.0357 # miss rate (i.e., misses/ref)
il1.repl_rate           0.0357 # replacement rate (i.e., repls/ref)
il1.wb_rate             0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate            0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses            217379569 # total number of accesses
dl1.hits                200332529 # total number of hits
dl1.misses              17047040 # total number of misses
dl1.replacements        17046528 # total number of replacements
dl1.writebacks          16837673 # total number of writebacks
dl1.invalidations       0 # total number of invalidations
dl1.miss_rate           0.0784 # miss rate (i.e., misses/ref)
dl1.repl_rate           0.0784 # replacement rate (i.e., repls/ref)
dl1.wb_rate             0.0775 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate            0.0000 # invalidation rate (i.e., invs/ref)
```

```
ul2.accesses              53897065 # total number of accesses
ul2.hits                  45518635 # total number of hits
ul2.misses                 8378430 # total number of misses
ul2.replacements           8374334 # total number of replacements
ul2.writebacks             8340774 # total number of writebacks
ul2.invalidations                0 # total number of invalidations
ul2.miss_rate               0.1555 # miss rate (i.e., misses/ref)
ul2.repl_rate               0.1554 # replacement rate (i.e., repls/ref)
ul2.wb_rate                 0.1548 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate                0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses             560197796 # total number of accesses
itlb.hits                 560163338 # total number of hits
itlb.misses                   34458 # total number of misses
itlb.replacements             34394 # total number of replacements
itlb.writebacks                   0 # total number of writebacks
itlb.invalidations                0 # total number of invalidations
itlb.miss_rate              0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate              0.0001 # replacement rate (i.e., repls/ref)
itlb.wb_rate                0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses             223942083 # total number of accesses
dtlb.hits                 223771080 # total number of hits
dtlb.misses                  171003 # total number of misses
dtlb.replacements            170875 # total number of replacements
dtlb.writebacks                   0 # total number of writebacks
dtlb.invalidations                0 # total number of invalidations
dtlb.miss_rate              0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate              0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate                0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs                 0 # total non-speculative bogus addresses seen (debug var)
ld_text_base             0x00400000 # program text (code) segment base
ld_text_size                 452000 # program text (code) size in bytes
ld_data_base             0x10000000 # program initialized data segment base
ld_data_size                2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base            0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size                16384 # program initial stack size
ld_prog_entry            0x00400140 # program entry point (initial PC)
ld_environ_base          0x7fff8000 # program environment base address address
ld_target_big_endian              0 # target executable endian-ness, non-zero if big endian
mem.page_count                45075 # total number of pages allocated
mem.page_mem                180300k # total size of memory pages allocated
mem.ptab_misses               93450 # total first level page table misses
mem.ptab_accesses        4013427964 # total page table accesses
mem.ptab_miss_rate          0.0000 # first level page table miss rate
```

Taken - ./sim-outorder -max:inst 500000000 -bpred taken -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000003 # total number of instructions committed
sim_num_refs            219903439 # total number of loads and stores committed
sim_num_loads            67608642 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time              212 # total simulation time in seconds
sim_inst_rate        2358490.5802 # simulation speed (in insts/sec)
sim_total_insn          889255891 # total number of instructions executed
sim_total_refs          313833617 # total number of loads and stores executed
sim_total_loads         114562260 # total number of loads executed
sim_total_stores      199271357.0000 # total number of stores executed
sim_total_branches      130393926 # total number of branches executed
sim_cycle             1065648875 # total simulation time in cycles
sim_IPC                    0.4692 # instructions per cycle
sim_CPI                    2.1313 # cycles per instruction
sim_exec_BW                0.8345 # total instructions (mis-spec + committed) per cycle
sim_IPB                    9.8578 # instruction per branch
IFQ_count             1310730988 # cumulative IFQ occupancy
IFQ_fcount             324602763 # cumulative IFQ full count
ifq_occupancy              1.2300 # avg IFQ occupancy (insn's)
ifq_rate                   0.8345 # avg IFQ dispatch rate (insn/cycle)
ifq_latency                1.4740 # avg IFQ occupant latency (cycle's)
ifq_full                   0.3046 # fraction of time (cycle's) IFQ was full
RUU_count             3760961529 # cumulative RUU occupancy
RUU_fcount             100667547 # cumulative RUU full count
ruu_occupancy              3.5293 # avg RUU occupancy (insn's)
ruu_rate                   0.8345 # avg RUU dispatch rate (insn/cycle)
ruu_latency                4.2293 # avg RUU occupant latency (cycle's)
ruu_full                   0.0945 # fraction of time (cycle's) RUU was full
LSQ_count             1374959799 # cumulative LSQ occupancy
LSQ_fcount             48710398 # cumulative LSQ full count
lsq_occupancy              1.2903 # avg LSQ occupancy (insn's)
lsq_rate                   0.8345 # avg LSQ dispatch rate (insn/cycle)
lsq_latency                1.5462 # avg LSQ occupant latency (cycle's)
lsq_full                   0.0457 # fraction of time (cycle's) LSQ was full
sim_slip              4565357238 # total number of slip cycles
avg_sim_slip               9.1307 # the average slip between issue and retirement
bpred_taken.lookups       172428034 # total number of bpred lookups
bpred_taken.updates        50721278 # total number of updates
bpred_taken.addr_hits       9548872 # total number of address-predicted hits
bpred_taken.dir_hits        9548872 # total number of direction-predicted hits (includes addr-hits)
bpred_taken.misses         41172406 # total number of misses
```

```
bpred_taken.jr_hits          109353 # total number of address-predicted hits for JR's
bpred_taken.jr_seen          1528398 # total number of JR's seen
bpred_taken.jr_non_ras_hits.PP       109353 # total number of address-predicted hits for non-RAS JR's
bpred_taken.jr_non_ras_seen.PP       1528398 # total number of non-RAS JR's seen
bpred_taken.bpred_addr_rate    0.1883 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_taken.bpred_dir_rate    0.1883 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_taken.bpred_jr_rate    0.0715 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_taken.bpred_jr_non_ras_rate.PP    0.0715 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_taken.retstack_pushes           0 # total number of address pushed onto ret-addr stack
bpred_taken.retstack_pops           0 # total number of address popped off of ret-addr stack
bpred_taken.used_ras.PP           0 # total number of RAS predictions used
bpred_taken.ras_hits.PP           0 # total number of RAS hits
bpred_taken.ras_rate.PP <error: divide by zero> # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            1054582579 # total number of accesses
il1.hits                1030925208 # total number of hits
il1.misses               23657371 # total number of misses
il1.replacements          23656859 # total number of replacements
il1.writebacks                 0 # total number of writebacks
il1.invalidations              0 # total number of invalidations
il1.miss_rate             0.0224 # miss rate (i.e., misses/ref)
il1.repl_rate             0.0224 # replacement rate (i.e., repls/ref)
il1.wb_rate               0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses            235109620 # total number of accesses
dl1.hits                218062296 # total number of hits
dl1.misses               17047324 # total number of misses
dl1.replacements          17046812 # total number of replacements
dl1.writebacks            16837555 # total number of writebacks
dl1.invalidations              0 # total number of invalidations
dl1.miss_rate             0.0725 # miss rate (i.e., misses/ref)
dl1.repl_rate             0.0725 # replacement rate (i.e., repls/ref)
dl1.wb_rate               0.0716 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses             57542250 # total number of accesses
ul2.hits                 49160881 # total number of hits
ul2.misses                8381369 # total number of misses
ul2.replacements           8377273 # total number of replacements
ul2.writebacks             8341314 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate             0.1457 # miss rate (i.e., misses/ref)
ul2.repl_rate             0.1456 # replacement rate (i.e., repls/ref)
ul2.wb_rate               0.1450 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses           1054582579 # total number of accesses
itlb.hits               1054547856 # total number of hits
itlb.misses                34723 # total number of misses
itlb.replacements          34659 # total number of replacements
itlb.writebacks                0 # total number of writebacks
itlb.invalidations             0 # total number of invalidations
```

```
itlb.miss_rate              0.0000 # miss rate (i.e., misses/ref)
itlb.repl_rate              0.0000 # replacement rate (i.e., repls/ref)
itlb.wb_rate                0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses           237914789 # total number of accesses
dtlb.hits               237743992 # total number of hits
dtlb.misses                170797 # total number of misses
dtlb.replacements          170669 # total number of replacements
dtlb.writebacks                 0 # total number of writebacks
dtlb.invalidations              0 # total number of invalidations
dtlb.miss_rate              0.0007 # miss rate (i.e., misses/ref)
dtlb.repl_rate              0.0007 # replacement rate (i.e., repls/ref)
dtlb.wb_rate                0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs               0 # total non-speculative bogus addresses seen (debug var)
ld_text_base            0x00400000 # program text (code) segment base
ld_text_size               452000 # program text (code) size in bytes
ld_data_base            0x10000000 # program initialized data segment base
ld_data_size              2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base           0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size              16384 # program initial stack size
ld_prog_entry           0x00400140 # program entry point (initial PC)
ld_environ_base         0x7fff8000 # program environment base address address
ld_target_big_endian            0 # target executable endian-ness, non-zero if big endian
mem.page_count              45075 # total number of pages allocated
mem.page_mem               180300k # total size of memory pages allocated
mem.ptab_misses             91868 # total first level page table misses
mem.ptab_accesses        6069208582 # total page table accesses
mem.ptab_miss_rate          0.0000 # first level page table miss rate
```

Not Taken - ./sim-outorder -max:inst 500000000 -bpred nottaken -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000003 # total number of instructions committed
sim_num_refs            219903439 # total number of loads and stores committed
sim_num_loads            67608642 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches          50721278 # total number of branches committed
sim_elapsed_time               201 # total simulation time in seconds
sim_inst_rate         2487562.2040 # simulation speed (in insts/sec)
sim_total_insn           829542101 # total number of instructions executed
sim_total_refs           294414920 # total number of loads and stores executed
sim_total_loads          111630369 # total number of loads executed
sim_total_stores      182784551.0000 # total number of stores executed
```

```
sim_total_branches        127253130 # total number of branches executed
sim_cycle                1069440418 # total simulation time in cycles
sim_IPC                     0.4675 # instructions per cycle
sim_CPI                     2.1389 # cycles per instruction
sim_exec_BW                 0.7757 # total instructions (mis-spec + committed) per cycle
sim_IPB                     9.8578 # instruction per branch
IFQ_count                1231530645 # cumulative IFQ occupancy
IFQ_fcount                267907463 # cumulative IFQ full count
ifq_occupancy               1.1516 # avg IFQ occupancy (insn's)
ifq_rate                    0.7757 # avg IFQ dispatch rate (insn/cycle)
ifq_latency                 1.4846 # avg IFQ occupant latency (cycle's)
ifq_full                    0.2505 # fraction of time (cycle's) IFQ was full
RUU_count                3632429802 # cumulative RUU occupancy
RUU_fcount                 77053846 # cumulative RUU full count
ruu_occupancy               3.3966 # avg RUU occupancy (insn's)
ruu_rate                    0.7757 # avg RUU dispatch rate (insn/cycle)
ruu_latency                 4.3788 # avg RUU occupant latency (cycle's)
ruu_full                    0.0721 # fraction of time (cycle's) RUU was full
LSQ_count                1326536087 # cumulative LSQ occupancy
LSQ_fcount                 48670499 # cumulative LSQ full count
lsq_occupancy               1.2404 # avg LSQ occupancy (insn's)
lsq_rate                    0.7757 # avg LSQ dispatch rate (insn/cycle)
lsq_latency                 1.5991 # avg LSQ occupant latency (cycle's)
lsq_full                    0.0455 # fraction of time (cycle's) LSQ was full
sim_slip                 4557313124 # total number of slip cycles
avg_sim_slip                9.1146 # the average slip between issue and retirement
bpred_nottaken.lookups    153089247 # total number of bpred lookups
bpred_nottaken.updates     50721278 # total number of updates
bpred_nottaken.addr_hits    9548872 # total number of address-predicted hits
bpred_nottaken.dir_hits     9548872 # total number of direction-predicted hits (includes addr-hits)
bpred_nottaken.misses      41172406 # total number of misses
bpred_nottaken.jr_hits       109353 # total number of address-predicted hits for JR's
bpred_nottaken.jr_seen      1528398 # total number of JR's seen
bpred_nottaken.jr_non_ras_hits.PP       109353 # total number of address-predicted hits for non-RAS JR's
bpred_nottaken.jr_non_ras_seen.PP      1528398 # total number of non-RAS JR's seen
bpred_nottaken.bpred_addr_rate    0.1883 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_nottaken.bpred_dir_rate    0.1883 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_nottaken.bpred_jr_rate    0.0715 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_nottaken.bpred_jr_non_ras_rate.PP    0.0715 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_nottaken.retstack_pushes           0 # total number of address pushed onto ret-addr stack
bpred_nottaken.retstack_pops             0 # total number of address popped off of ret-addr stack
bpred_nottaken.used_ras.PP               0 # total number of RAS predictions used
bpred_nottaken.ras_hits.PP               0 # total number of RAS hits
bpred_nottaken.ras_rate.PP <error: divide by zero> # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses             991718805 # total number of accesses
il1.hits                 968206077 # total number of hits
il1.misses                23512728 # total number of misses
il1.replacements          23512216 # total number of replacements
il1.writebacks                   0 # total number of writebacks
```

```
il1.invalidations              0 # total number of invalidations
il1.miss_rate             0.0237 # miss rate (i.e., misses/ref)
il1.repl_rate             0.0237 # replacement rate (i.e., repls/ref)
il1.wb_rate               0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses           234194305 # total number of accesses
dl1.hits               217147045 # total number of hits
dl1.misses              17047260 # total number of misses
dl1.replacements        17046748 # total number of replacements
dl1.writebacks          16837692 # total number of writebacks
dl1.invalidations              0 # total number of invalidations
dl1.miss_rate             0.0728 # miss rate (i.e., misses/ref)
dl1.repl_rate             0.0728 # replacement rate (i.e., repls/ref)
dl1.wb_rate               0.0719 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses            57397680 # total number of accesses
ul2.hits                49016427 # total number of hits
ul2.misses               8381253 # total number of misses
ul2.replacements         8377157 # total number of replacements
ul2.writebacks           8341295 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate             0.1460 # miss rate (i.e., misses/ref)
ul2.repl_rate             0.1459 # replacement rate (i.e., repls/ref)
ul2.wb_rate               0.1453 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses          991718805 # total number of accesses
itlb.hits              991684082 # total number of hits
itlb.misses                34723 # total number of misses
itlb.replacements          34659 # total number of replacements
itlb.writebacks                0 # total number of writebacks
itlb.invalidations             0 # total number of invalidations
itlb.miss_rate            0.0000 # miss rate (i.e., misses/ref)
itlb.repl_rate            0.0000 # replacement rate (i.e., repls/ref)
itlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses          236990573 # total number of accesses
dtlb.hits              236819776 # total number of hits
dtlb.misses               170797 # total number of misses
dtlb.replacements         170669 # total number of replacements
dtlb.writebacks                0 # total number of writebacks
dtlb.invalidations             0 # total number of invalidations
dtlb.miss_rate            0.0007 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0007 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs              0 # total non-speculative bogus addresses seen (debug var)
ld_text_base          0x00400000 # program text (code) segment base
ld_text_size              452000 # program text (code) size in bytes
ld_data_base          0x10000000 # program initialized data segment base
```

```
ld_data_size            2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base         0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size             16384 # program initial stack size
ld_prog_entry         0x00400140 # program entry point (initial PC)
ld_environ_base        0x7fff8000 # program environment base address address
ld_target_big_endian           0 # target executable endian-ness, non-zero if big endian
mem.page_count            45075 # total number of pages allocated
mem.page_mem             180300k # total size of memory pages allocated
mem.ptab_misses           91868 # total first level page table misses
mem.ptab_accesses      5811782198 # total page table accesses
mem.ptab_miss_rate        0.0000 # first level page table miss rate
```

Comb - ./sim-outorder -max:inst 500000000 -bpred comb -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000002 # total number of instructions committed
sim_num_refs            219903438 # total number of loads and stores committed
sim_num_loads            67608641 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time               154 # total simulation time in seconds
sim_inst_rate        3246753.2597 # simulation speed (in insts/sec)
sim_total_insn          525357673 # total number of instructions executed
sim_total_refs          226581451 # total number of loads and stores executed
sim_total_loads          72848326 # total number of loads executed
sim_total_stores      153733125.0000 # total number of stores executed
sim_total_branches       54438971 # total number of branches executed
sim_cycle               397966523 # total simulation time in cycles
sim_IPC                    1.2564 # instructions per cycle
sim_CPI                    0.7959 # cycles per instruction
sim_exec_BW                1.3201 # total instructions (mis-spec + committed) per cycle
sim_IPB                    9.8578 # instruction per branch
IFQ_count               850217259 # cumulative IFQ occupancy
IFQ_fcount              201202809 # cumulative IFQ full count
ifq_occupancy              2.1364 # avg IFQ occupancy (insn's)
ifq_rate                   1.3201 # avg IFQ dispatch rate (insn/cycle)
ifq_latency                1.6184 # avg IFQ occupant latency (cycle's)
ifq_full                   0.5056 # fraction of time (cycle's) IFQ was full
RUU_count              3051729933 # cumulative RUU occupancy
RUU_fcount               85309459 # cumulative RUU full count
ruu_occupancy              7.6683 # avg RUU occupancy (insn's)
ruu_rate                   1.3201 # avg RUU dispatch rate (insn/cycle)
ruu_latency                5.8089 # avg RUU occupant latency (cycle's)
ruu_full                   0.2144 # fraction of time (cycle's) RUU was full
```

```
LSQ_count              1256483263 # cumulative LSQ occupancy
LSQ_fcount               72936094 # cumulative LSQ full count
lsq_occupancy              3.1573 # avg LSQ occupancy (insn's)
lsq_rate                   1.3201 # avg LSQ dispatch rate (insn/cycle)
lsq_latency                2.3917 # avg LSQ occupant latency (cycle's)
lsq_full                   0.1833 # fraction of time (cycle's) LSQ was full
sim_slip               4891683362 # total number of slip cycles
avg_sim_slip               9.7834 # the average slip between issue and retirement
bpred_comb.lookups         55049393 # total number of bpred lookups
bpred_comb.updates         50721275 # total number of updates
bpred_comb.addr_hits       46665818 # total number of address-predicted hits
bpred_comb.dir_hits        47090104 # total number of direction-predicted hits (includes addr-hits)
bpred_comb.used_bimod      33928213 # total number of bimodal predictions used
bpred_comb.used_2lev        4838553 # total number of 2-level predictions used
bpred_comb.misses           3631171 # total number of misses
bpred_comb.jr_hits          1116140 # total number of address-predicted hits for JR's
bpred_comb.jr_seen          1528398 # total number of JR's seen
bpred_comb.jr_non_ras_hits.PP       106459 # total number of address-predicted hits for non-RAS JR's
bpred_comb.jr_non_ras_seen.PP       389523 # total number of non-RAS JR's seen
bpred_comb.bpred_addr_rate    0.9200 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_comb.bpred_dir_rate    0.9284 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_comb.bpred_jr_rate    0.7303 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_comb.bpred_jr_non_ras_rate.PP    0.2733 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_comb.retstack_pushes      1290232 # total number of address pushed onto ret-addr stack
bpred_comb.retstack_pops        1190953 # total number of address popped off of ret-addr stack
bpred_comb.used_ras.PP          1138875 # total number of RAS predictions used
bpred_comb.ras_hits.PP          1009681 # total number of RAS hits
bpred_comb.ras_rate.PP    0.8866 # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            552925090 # total number of accesses
il1.hits                533680680 # total number of hits
il1.misses               19244410 # total number of misses
il1.replacements         19243898 # total number of replacements
il1.writebacks                  0 # total number of writebacks
il1.invalidations               0 # total number of invalidations
il1.miss_rate              0.0348 # miss rate (i.e., misses/ref)
il1.repl_rate              0.0348 # replacement rate (i.e., repls/ref)
il1.wb_rate                0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses            216044807 # total number of accesses
dl1.hits                198998196 # total number of hits
dl1.misses               17046611 # total number of misses
dl1.replacements         17046099 # total number of replacements
dl1.writebacks           16837641 # total number of writebacks
dl1.invalidations               0 # total number of invalidations
dl1.miss_rate              0.0789 # miss rate (i.e., misses/ref)
dl1.repl_rate              0.0789 # replacement rate (i.e., repls/ref)
dl1.wb_rate                0.0779 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses             53128662 # total number of accesses
```

```
ul2.hits                44751445 # total number of hits
ul2.misses               8377217 # total number of misses
ul2.replacements         8373121 # total number of replacements
ul2.writebacks           8340504 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate             0.1577 # miss rate (i.e., misses/ref)
ul2.repl_rate             0.1576 # replacement rate (i.e., repls/ref)
ul2.wb_rate               0.1570 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses           552925090 # total number of accesses
itlb.hits               552890699 # total number of hits
itlb.misses                 34391 # total number of misses
itlb.replacements           34327 # total number of replacements
itlb.writebacks                 0 # total number of writebacks
itlb.invalidations              0 # total number of invalidations
itlb.miss_rate            0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate            0.0001 # replacement rate (i.e., repls/ref)
itlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses           222994237 # total number of accesses
dtlb.hits               222823222 # total number of hits
dtlb.misses                171015 # total number of misses
dtlb.replacements          170887 # total number of replacements
dtlb.writebacks                 0 # total number of writebacks
dtlb.invalidations              0 # total number of invalidations
dtlb.miss_rate            0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs               0 # total non-speculative bogus addresses seen (debug var)
ld_text_base           0x00400000 # program text (code) segment base
ld_text_size              452000 # program text (code) size in bytes
ld_data_base           0x10000000 # program initialized data segment base
ld_data_size             2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base          0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size              16384 # program initial stack size
ld_prog_entry          0x00400140 # program entry point (initial PC)
ld_environ_base        0x7fff8000 # program environment base address address
ld_target_big_endian            0 # target executable endian-ness, non-zero if big endian
mem.page_count              45075 # total number of pages allocated
mem.page_mem               180300k # total size of memory pages allocated
mem.ptab_misses             93452 # total first level page table misses
mem.ptab_accesses        3981485675 # total page table accesses
mem.ptab_miss_rate          0.0000 # first level page table miss rate
```

Bimod -./sim-outorder -max:inst 500000000 -bpred bimod -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000002 # total number of instructions committed
sim_num_refs            219903438 # total number of loads and stores committed
sim_num_loads            67608641 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time              139 # total simulation time in seconds
sim_inst_rate        3597122.3165 # simulation speed (in insts/sec)
sim_total_insn          529924027 # total number of instructions executed
sim_total_refs          227552946 # total number of loads and stores executed
sim_total_loads          73679940 # total number of loads executed
sim_total_stores      153873006.0000 # total number of stores executed
sim_total_branches       55047853 # total number of branches executed
sim_cycle               407108272 # total simulation time in cycles
sim_IPC                    1.2282 # instructions per cycle
sim_CPI                    0.8142 # cycles per instruction
sim_exec_BW                1.3017 # total instructions (mis-spec + committed) per cycle
sim_IPB                    9.8578 # instruction per branch
IFQ_count               859145502 # cumulative IFQ occupancy
IFQ_fcount              203240104 # cumulative IFQ full count
ifq_occupancy              2.1104 # avg IFQ occupancy (insn's)
ifq_rate                   1.3017 # avg IFQ dispatch rate (insn/cycle)
ifq_latency                1.6213 # avg IFQ occupant latency (cycle's)
ifq_full                   0.4992 # fraction of time (cycle's) IFQ was full
RUU_count              3068953622 # cumulative RUU occupancy
RUU_fcount              86237430 # cumulative RUU full count
ruu_occupancy              7.5384 # avg RUU occupancy (insn's)
ruu_rate                   1.3017 # avg RUU dispatch rate (insn/cycle)
ruu_latency                5.7913 # avg RUU occupant latency (cycle's)
ruu_full                   0.2118 # fraction of time (cycle's) RUU was full
LSQ_count              1260035216 # cumulative LSQ occupancy
LSQ_fcount              72883800 # cumulative LSQ full count
lsq_occupancy              3.0951 # avg LSQ occupancy (insn's)
lsq_rate                   1.3017 # avg LSQ dispatch rate (insn/cycle)
lsq_latency                2.3778 # avg LSQ occupant latency (cycle's)
lsq_full                   0.1790 # fraction of time (cycle's) LSQ was full
sim_slip               4887886392 # total number of slip cycles
avg_sim_slip               9.7758 # the average slip between issue and retirement
bpred_bimod.lookups        55713732 # total number of bpred lookups
bpred_bimod.updates        50721275 # total number of updates
bpred_bimod.addr_hits      46203927 # total number of address-predicted hits
bpred_bimod.dir_hits       46632354 # total number of direction-predicted hits (includes addr-hits)
bpred_bimod.misses          4088921 # total number of misses
bpred_bimod.jr_hits         1111886 # total number of address-predicted hits for JR's
bpred_bimod.jr_seen         1528398 # total number of JR's seen
```

```
bpred_bimod.jr_non_ras_hits.PP       106459 # total number of address-predicted hits for non-RAS JR's
bpred_bimod.jr_non_ras_seen.PP       389523 # total number of non-RAS JR's seen
bpred_bimod.bpred_addr_rate    0.9109 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_bimod.bpred_dir_rate    0.9194 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_bimod.bpred_jr_rate    0.7275 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_bimod.bpred_jr_non_ras_rate.PP    0.2733 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_bimod.retstack_pushes      1310186 # total number of address pushed onto ret-addr stack
bpred_bimod.retstack_pops      1195499 # total number of address popped off of ret-addr stack
bpred_bimod.used_ras.PP      1138875 # total number of RAS predictions used
bpred_bimod.ras_hits.PP      1005427 # total number of RAS hits
bpred_bimod.ras_rate.PP    0.8828 # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            558908281 # total number of accesses
il1.hits               539687834 # total number of hits
il1.misses              19220447 # total number of misses
il1.replacements          19219935 # total number of replacements
il1.writebacks                 0 # total number of writebacks
il1.invalidations              0 # total number of invalidations
il1.miss_rate            0.0344 # miss rate (i.e., misses/ref)
il1.repl_rate            0.0344 # replacement rate (i.e., repls/ref)
il1.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses            216466726 # total number of accesses
dl1.hits               199420063 # total number of hits
dl1.misses              17046663 # total number of misses
dl1.replacements          17046151 # total number of replacements
dl1.writebacks           16837654 # total number of writebacks
dl1.invalidations              0 # total number of invalidations
dl1.miss_rate            0.0787 # miss rate (i.e., misses/ref)
dl1.repl_rate            0.0787 # replacement rate (i.e., repls/ref)
dl1.wb_rate              0.0778 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses            53104764 # total number of accesses
ul2.hits               44727754 # total number of hits
ul2.misses              8377010 # total number of misses
ul2.replacements          8372914 # total number of replacements
ul2.writebacks           8340468 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate            0.1577 # miss rate (i.e., misses/ref)
ul2.repl_rate            0.1577 # replacement rate (i.e., repls/ref)
ul2.wb_rate              0.1571 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses           558908281 # total number of accesses
itlb.hits              558873542 # total number of hits
itlb.misses                34739 # total number of misses
itlb.replacements          34675 # total number of replacements
itlb.writebacks                0 # total number of writebacks
itlb.invalidations             0 # total number of invalidations
itlb.miss_rate           0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate           0.0001 # replacement rate (i.e., repls/ref)
```

```
itlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses           223485974 # total number of accesses
dtlb.hits               223314967 # total number of hits
dtlb.misses               171007 # total number of misses
dtlb.replacements         170879 # total number of replacements
dtlb.writebacks                0 # total number of writebacks
dtlb.invalidations             0 # total number of invalidations
dtlb.miss_rate            0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs              0 # total non-speculative bogus addresses seen (debug var)
ld_text_base           0x00400000 # program text (code) segment base
ld_text_size              452000 # program text (code) size in bytes
ld_data_base           0x10000000 # program initialized data segment base
ld_data_size             2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base          0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size              16384 # program initial stack size
ld_prog_entry          0x00400140 # program entry point (initial PC)
ld_environ_base        0x7fff8000 # program environment base address address
ld_target_big_endian           0 # target executable endian-ness, non-zero if big endian
mem.page_count             45075 # total number of pages allocated
mem.page_mem              180300k # total size of memory pages allocated
mem.ptab_misses            93446 # total first level page table misses
mem.ptab_accesses       4006959950 # total page table accesses
mem.ptab_miss_rate         0.0000 # first level page table miss rate
```

2lev - ./sim-outorder -max:inst 500000000 -bpred 2lev -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000002 # total number of instructions committed
sim_num_refs            219903438 # total number of loads and stores committed
sim_num_loads            67608641 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time              128 # total simulation time in seconds
sim_inst_rate         3906250.0156 # simulation speed (in insts/sec)
sim_total_insn           536421403 # total number of instructions executed
sim_total_refs           230051613 # total number of loads and stores executed
sim_total_loads           75450977 # total number of loads executed
sim_total_stores       154600636.0000 # total number of stores executed
sim_total_branches        57378908 # total number of branches executed
sim_cycle              443172419 # total simulation time in cycles
sim_IPC                    1.1282 # instructions per cycle
```

```
sim_CPI                 0.8863 # cycles per instruction
sim_exec_BW             1.2104 # total instructions (mis-spec + committed) per cycle
sim_IPB                 9.8578 # instruction per branch
IFQ_count               855868484 # cumulative IFQ occupancy
IFQ_fcount              200478527 # cumulative IFQ full count
ifq_occupancy           1.9312 # avg IFQ occupancy (insn's)
ifq_rate                1.2104 # avg IFQ dispatch rate (insn/cycle)
ifq_latency             1.5955 # avg IFQ occupant latency (cycle's)
ifq_full                0.4524 # fraction of time (cycle's) IFQ was full
RUU_count               3079007298 # cumulative RUU occupancy
RUU_fcount              81595562 # cumulative RUU full count
ruu_occupancy           6.9477 # avg RUU occupancy (insn's)
ruu_rate                1.2104 # avg RUU dispatch rate (insn/cycle)
ruu_latency             5.7399 # avg RUU occupant latency (cycle's)
ruu_full                0.1841 # fraction of time (cycle's) RUU was full
LSQ_count               1263769533 # cumulative LSQ occupancy
LSQ_fcount              72066400 # cumulative LSQ full count
lsq_occupancy           2.8516 # avg LSQ occupancy (insn's)
lsq_rate                1.2104 # avg LSQ dispatch rate (insn/cycle)
lsq_latency             2.3559 # avg LSQ occupant latency (cycle's)
lsq_full                0.1626 # fraction of time (cycle's) LSQ was full
sim_slip                4869453514 # total number of slip cycles
avg_sim_slip            9.7389 # the average slip between issue and retirement
bpred_2lev.lookups      58425119 # total number of bpred lookups
bpred_2lev.updates      50721275 # total number of updates
bpred_2lev.addr_hits    44084488 # total number of address-predicted hits
bpred_2lev.dir_hits     44503081 # total number of direction-predicted hits (includes addr-hits)
bpred_2lev.misses       6218194 # total number of misses
bpred_2lev.jr_hits      1120350 # total number of address-predicted hits for JR's
bpred_2lev.jr_seen      1528398 # total number of JR's seen
bpred_2lev.jr_non_ras_hits.PP     106459 # total number of address-predicted hits for non-RAS JR's
bpred_2lev.jr_non_ras_seen.PP     389523 # total number of non-RAS JR's seen
bpred_2lev.bpred_addr_rate    0.8692 # branch address-prediction rate (i.e., addr-hits/updates)
bpred_2lev.bpred_dir_rate     0.8774 # branch direction-prediction rate (i.e., all-hits/updates)
bpred_2lev.bpred_jr_rate      0.7330 # JR address-prediction rate (i.e., JR addr-hits/JRs seen)
bpred_2lev.bpred_jr_non_ras_rate.PP    0.2733 # non-RAS JR addr-pred rate (ie, non-RAS JR hits/JRs seen)
bpred_2lev.retstack_pushes    1355043 # total number of address pushed onto ret-addr stack
bpred_2lev.retstack_pops      1294956 # total number of address popped off of ret-addr stack
bpred_2lev.used_ras.PP        1138875 # total number of RAS predictions used
bpred_2lev.ras_hits.PP        1013891 # total number of RAS hits
bpred_2lev.ras_rate.PP        0.8903 # RAS prediction rate (i.e., RAS hits/used RAS)
il1.accesses            567205719 # total number of accesses
il1.hits                547003864 # total number of hits
il1.misses              20201855 # total number of misses
il1.replacements        20201343 # total number of replacements
il1.writebacks          0 # total number of writebacks
il1.invalidations       0 # total number of invalidations
il1.miss_rate           0.0356 # miss rate (i.e., misses/ref)
il1.repl_rate           0.0356 # replacement rate (i.e., repls/ref)
```

```
il1.wb_rate                0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate               0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses           218531254 # total number of accesses
dl1.hits               201484572 # total number of hits
dl1.misses              17046682 # total number of misses
dl1.replacements        17046170 # total number of replacements
dl1.writebacks          16837649 # total number of writebacks
dl1.invalidations              0 # total number of invalidations
dl1.miss_rate             0.0780 # miss rate (i.e., misses/ref)
dl1.repl_rate             0.0780 # replacement rate (i.e., repls/ref)
dl1.wb_rate               0.0770 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses            54086186 # total number of accesses
ul2.hits                45708070 # total number of hits
ul2.misses               8378116 # total number of misses
ul2.replacements         8374020 # total number of replacements
ul2.writebacks           8340708 # total number of writebacks
ul2.invalidations              0 # total number of invalidations
ul2.miss_rate             0.1549 # miss rate (i.e., misses/ref)
ul2.repl_rate             0.1548 # replacement rate (i.e., repls/ref)
ul2.wb_rate               0.1542 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate              0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses          567205719 # total number of accesses
itlb.hits              567171575 # total number of hits
itlb.misses               34144 # total number of misses
itlb.replacements         34080 # total number of replacements
itlb.writebacks               0 # total number of writebacks
itlb.invalidations            0 # total number of invalidations
itlb.miss_rate            0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate            0.0001 # replacement rate (i.e., repls/ref)
itlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses          224665473 # total number of accesses
dtlb.hits              224494443 # total number of hits
dtlb.misses               171030 # total number of misses
dtlb.replacements         170902 # total number of replacements
dtlb.writebacks               0 # total number of writebacks
dtlb.invalidations            0 # total number of invalidations
dtlb.miss_rate            0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs              0 # total non-speculative bogus addresses seen (debug var)
ld_text_base          0x00400000 # program text (code) segment base
ld_text_size              452000 # program text (code) size in bytes
ld_data_base          0x10000000 # program initialized data segment base
ld_data_size             2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base         0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size             16384 # program initial stack size
```

```
ld_prog_entry          0x00400140 # program entry point (initial PC)
ld_environ_base        0x7fff8000 # program environment base address address
ld_target_big_endian           0 # target executable endian-ness, non-zero if big endian
mem.page_count             45075 # total number of pages allocated
mem.page_mem              180300k # total size of memory pages allocated
mem.ptab_misses            93475 # total first level page table misses
mem.ptab_accesses       4043870626 # total page table accesses
mem.ptab_miss_rate         0.0000 # first level page table miss rate
```

Perfect - ./sim-outorder -max:inst 500000000 -bpred perfect -fetch:mplat 15 sjeng reduced.txt

```
sim: ** simulation statistics **
sim_num_insn            500000002 # total number of instructions committed
sim_num_refs            219903438 # total number of loads and stores committed
sim_num_loads            67608641 # total number of loads committed
sim_num_stores        152294797.0000 # total number of stores committed
sim_num_branches         50721278 # total number of branches committed
sim_elapsed_time              112 # total simulation time in seconds
sim_inst_rate        4464285.7321 # simulation speed (in insts/sec)
sim_total_insn          500000002 # total number of instructions executed
sim_total_refs          219903438 # total number of loads and stores executed
sim_total_loads          67608641 # total number of loads executed
sim_total_stores      152294797.0000 # total number of stores executed
sim_total_branches       50721278 # total number of branches executed
sim_cycle               327402153 # total simulation time in cycles
sim_IPC                    1.5272 # instructions per cycle
sim_CPI                    0.6548 # cycles per instruction
sim_exec_BW                1.5272 # total instructions (mis-spec + committed) per cycle
sim_IPB                    9.8578 # instruction per branch
IFQ_count               847131960 # cumulative IFQ occupancy
IFQ_fcount              209209254 # cumulative IFQ full count
ifq_occupancy              2.5874 # avg IFQ occupancy (insn's)
ifq_rate                   1.5272 # avg IFQ dispatch rate (insn/cycle)
ifq_latency                1.6943 # avg IFQ occupant latency (cycle's)
ifq_full                   0.6390 # fraction of time (cycle's) IFQ was full
RUU_count              2965542612 # cumulative RUU occupancy
RUU_fcount               80694860 # cumulative RUU full count
ruu_occupancy              9.0578 # avg RUU occupancy (insn's)
ruu_rate                   1.5272 # avg RUU dispatch rate (insn/cycle)
ruu_latency                5.9311 # avg RUU occupant latency (cycle's)
ruu_full                   0.2465 # fraction of time (cycle's) RUU was full
LSQ_count              1235188407 # cumulative LSQ occupancy
LSQ_fcount               72986531 # cumulative LSQ full count
lsq_occupancy              3.7727 # avg LSQ occupancy (insn's)
```

```
lsq_rate                1.5272 # avg LSQ dispatch rate (insn/cycle)
lsq_latency             2.4704 # avg LSQ occupant latency (cycle's)
lsq_full                0.2229 # fraction of time (cycle's) LSQ was full
sim_slip           4920300216 # total number of slip cycles
avg_sim_slip            9.8406 # the average slip between issue and retirement
il1.accesses        562954702 # total number of accesses
il1.hits            543464471 # total number of hits
il1.misses           19490231 # total number of misses
il1.replacements     19489719 # total number of replacements
il1.writebacks             0 # total number of writebacks
il1.invalidations          0 # total number of invalidations
il1.miss_rate           0.0346 # miss rate (i.e., misses/ref)
il1.repl_rate           0.0346 # replacement rate (i.e., repls/ref)
il1.wb_rate             0.0000 # writeback rate (i.e., wrbks/ref)
il1.inv_rate            0.0000 # invalidation rate (i.e., invs/ref)
dl1.accesses        213401941 # total number of accesses
dl1.hits            196355771 # total number of hits
dl1.misses           17046170 # total number of misses
dl1.replacements     17045658 # total number of replacements
dl1.writebacks       16837647 # total number of writebacks
dl1.invalidations          0 # total number of invalidations
dl1.miss_rate           0.0799 # miss rate (i.e., misses/ref)
dl1.repl_rate           0.0799 # replacement rate (i.e., repls/ref)
dl1.wb_rate             0.0789 # writeback rate (i.e., wrbks/ref)
dl1.inv_rate            0.0000 # invalidation rate (i.e., invs/ref)
ul2.accesses         53374048 # total number of accesses
ul2.hits             44996809 # total number of hits
ul2.misses            8377239 # total number of misses
ul2.replacements      8373143 # total number of replacements
ul2.writebacks        8340602 # total number of writebacks
ul2.invalidations          0 # total number of invalidations
ul2.miss_rate           0.1570 # miss rate (i.e., misses/ref)
ul2.repl_rate           0.1569 # replacement rate (i.e., repls/ref)
ul2.wb_rate             0.1563 # writeback rate (i.e., wrbks/ref)
ul2.inv_rate            0.0000 # invalidation rate (i.e., invs/ref)
itlb.accesses       562954702 # total number of accesses
itlb.hits           562920147 # total number of hits
itlb.misses             34555 # total number of misses
itlb.replacements       34491 # total number of replacements
itlb.writebacks            0 # total number of writebacks
itlb.invalidations         0 # total number of invalidations
itlb.miss_rate          0.0001 # miss rate (i.e., misses/ref)
itlb.repl_rate          0.0001 # replacement rate (i.e., repls/ref)
itlb.wb_rate            0.0000 # writeback rate (i.e., wrbks/ref)
itlb.inv_rate           0.0000 # invalidation rate (i.e., invs/ref)
dtlb.accesses       219903434 # total number of accesses
dtlb.hits           219732643 # total number of hits
dtlb.misses            170791 # total number of misses
dtlb.replacements      170663 # total number of replacements
```

```
dtlb.writebacks                0 # total number of writebacks
dtlb.invalidations             0 # total number of invalidations
dtlb.miss_rate            0.0008 # miss rate (i.e., misses/ref)
dtlb.repl_rate            0.0008 # replacement rate (i.e., repls/ref)
dtlb.wb_rate              0.0000 # writeback rate (i.e., wrbks/ref)
dtlb.inv_rate             0.0000 # invalidation rate (i.e., invs/ref)
sim_invalid_addrs              0 # total non-speculative bogus addresses seen (debug var)
ld_text_base           0x00400000 # program text (code) segment base
ld_text_size              452000 # program text (code) size in bytes
ld_data_base           0x10000000 # program initialized data segment base
ld_data_size             2615828 # program init'ed `.data' and uninit'ed `.bss' size in bytes
ld_stack_base          0x7fffc000 # program stack segment base (highest address in stack)
ld_stack_size             16384 # program initial stack size
ld_prog_entry          0x00400140 # program entry point (initial PC)
ld_environ_base        0x7fff8000 # program environment base address address
ld_target_big_endian           0 # target executable endian-ness, non-zero if big endian
mem.page_count            45075 # total number of pages allocated
mem.page_mem             180300k # total size of memory pages allocated
mem.ptab_misses           93385 # total first level page table misses
mem.ptab_accesses       4011974544 # total page table accesses
mem.ptab_miss_rate        0.0000 # first level page table miss rate
```